Tutorial IV.IV - Advanced Solver Options with HiGHS in JuMP

Applied Optimization with Julia



Introduction

Welcome to this tutorial on advanced solver options in JuMP using the HiGHS solver! Don't worry if "advanced solver options" sounds intimidating - we'll break everything down into simple, easy-to-understand concepts.

Imagine you're using a GPS app to find the best route to a new restaurant. Just like how you can adjust settings in your GPS (like avoiding toll roads or preferring highways), we can adjust settings in our optimization solver to help it find solutions more efficiently or to meet specific requirements.

By the end of this tutorial, you'll be able to:

1. Understand what solver options are and why they're useful
2. Set basic solver options like time limits and solution tolerances
3. Interpret solver output to understand how well your problem was solved

Let's start by loading the necessary packages:

```
using JuMP, HiGHS
```

Section 1 - Understanding Solver Options

Solver options are like the "advanced settings" of our optimization tool. They allow us to control how the solver approaches our problem. Here are a few common options:

1. **Time limit**: How long the solver should try before giving up
2. **Solution tolerance**: How precise we need the answer to be
3. **Presolve**: Whether to simplify the problem before solving it

Let's create a model and set some of these options:

```
model = Model(HiGHS.Optimizer)

# Set a time limit of 60 seconds
set_time_limit_sec(model, 60)

# Set the relative MIP gap tolerance to 1%
set_optimizer_attribute(model, "mip_rel_gap", 0.01)

# Turn on presolve
set_optimizer_attribute(model, "presolve", "on")
```

```
println("Solver options set successfully!")
```

```
Solver options set successfully!
```

Let's break this down:

- `set_time_limit_sec(model, 60)` tells the solver to stop after 60 seconds and return the best solution it has found so far, if any
- `set_optimizer_attribute(model, "mip_rel_gap", 0.01)` sets how close to the best possible solution we need to be (within 1%). This means the solver is allowed to stop as soon as it can prove that its best solution is at most 1% worse than the true optimum - even if it is not the exact optimum. We will see this in action in the next section.
- `set_optimizer_attribute(model, "presolve", "on")` tells the solver to try simplifying the problem first

Exercise 1.1 - Set Solver Options

Now it's your turn! Set the following solver options:

1. A time limit of 120 seconds
2. A MIP gap tolerance of 0.5%
3. Turn off presolve

```
# YOUR CODE BELOW
```

```
# Test your answer
@assert time_limit_sec(model) == 120 "The time limit should be 120 seconds but
is $(time_limit_sec(model))"
@assert solver_name(model) == "HiGHS" "The solver should be HiGHS but is
$(solver_name(model))"
@assert MOI.get(model, MOI.RawOptimizerAttribute("mip_rel_gap")) == 0.005 "The
MIP gap should be 0.5% but is $(MOI.get(model,
MOI.RawOptimizerAttribute("mip_rel_gap")))"
@assert MOI.get(model, MOI.RawOptimizerAttribute("presolve")) == "off"
"Presolve should be off but is $(MOI.get(model,
MOI.RawOptimizerAttribute("presolve")))"
println("Great job! You've successfully set advanced solver options.")
```

Section 2 - Creating and Solving a Sample Problem

To see how these options affect solving, let's create a simple optimization problem. We'll use a basic production planning scenario.

Imagine you're managing a small factory that produces two types of products: widgets and gadgets. You want to maximize profit while staying within your production capacity.

```
# Define variables
@variable(model, widgets >= 0, Int)
@variable(model, gadgets >= 0, Int)

# Define constraints
# We have 240 minutes of production time available
@constraint(model,
    production_time,
    2*widgets + 3*gadgets <= 240
)
@constraint(model,
    widget_demand,
    widgets <= 80
)
@constraint(model,
    gadget_demand,
    gadgets <= 60
)

# Define objective (profit)
@objective(model,
    Max,
    25*widgets + 30*gadgets
)

# The solver options from Section 1 still belong to this model!
# We tighten the MIP gap to 0, so the solver proves the exact optimum
set_optimizer_attribute(model, "mip_rel_gap", 0.0)

# Solve the problem
optimize!(model)

# Print results
println("Optimization status: ", termination_status(model))
println("Objective value: ", objective_value(model))
println("Widgets to produce: ", value(widgets))
println("Gadgets to produce: ", value(gadgets))
```

```
Running HiGHS 1.15.1 (git hash: 04024d701f): Copyright (c) 2026 under MIT
licence terms
Includes third-party software components, see THIRD_PARTY_NOTICES.md for full
details
Using BLAS: Apple Accelerate
MIP has 3 rows; 2 cols; 4 nonzeros; 2 integer variables (0 binary)
Coefficient ranges:
```

```
  Matrix  [1e+00, 3e+00]
  Cost    [2e+01, 3e+01]
  Bound   [0e+00, 0e+00]
  RHS     [6e+01, 2e+02]
Presolving model
1 rows, 2 cols, 2 nonzeros 0s
1 rows, 2 cols, 2 nonzeros 0s
Presolve reductions: rows 1(-2); columns 2(-0); nonzeros 2(-2)
Objective function is integral with scale 0.2

Solving MIP model with:
   1 row
   2 cols (0 binary, 2 integer, 0 implied int., 0 continuous, 0 domain fixed)
   2 nonzeros
   Thread count 5 (of 10 threads). Using 1 max workers. Parallel search off

Src: B => Branching; C => Central rounding; F => Feasibility pump; H =>
Heuristic;
     I => Shifting; J => Feasibility jump; L => Sub-MIP; P => Empty MIP; R =>
Randomized rounding;
     S => Solve LP; T => Evaluate node; U => Unbounded; X => User solution; Y
=> HiGHS solution;
     Z => ZI Round; l => Trivial lower; p => Trivial point; u => Trivial upper;
z => Trivial zero

        Nodes      |    B&B Tree     |            Objective Bounds
|  Dynamic Constraints |       Work
Src  Proc. InQueue |  Leaves   Expl. | BestBound       BestSol              Gap
|   Cuts   InLp Confl. | LpIters     Time

 l       0       0         0   0.00%   inf             1530                Large
0      0      0         0     0.0s
 J       0       0         0   0.00%   inf             2780                Large
0      0      0         0     0.0s
 S       0       0         0   0.00%   2815            2785                1.08%
0      0      0         0     0.0s
 T       0       0         0   0.00%   2795            2790                0.18%
0      0      1         0     0.0s
         1       0         1 100.00%   2790            2790                0.00%
0      0      1         0     0.0s

Solving report
  Status            Optimal
  Primal bound      2790
  Dual bound        2790
  Gap               0%
  P-D integral      1.73513955848e-06
  Solution status   feasible
```

```
                    2790 (objective)
                    0 (bound viol.)
                    0 (int. viol.)
                    0 (row viol.)
  Timing            0.00
                    0.00 (Presolve)
                    0.00 (Solve)
                    0.00 (Postsolve)
  Max sub-MIP depth 0
  Nodes             1
  Repair LPs        0
  LP iterations     0
Optimization status: OPTIMAL
Objective value: 2790.0
Widgets to produce: 78.0
Gadgets to produce: 28.0
```

This problem determines how many widgets and gadgets to produce to maximize profit, given the available production time in minutes and the maximum demand for each product.

Take a close look at the results. The best plan is 78 widgets and 28 gadgets, with a profit of 2790. Notice that the solver does not simply max out widgets, even though widgets earn the most profit per minute. The “greedy” plan - produce 80 widgets first, then fill the remaining time with 26 gadgets - earns only 2780. If we allowed fractional production, the best plan would be 80 widgets and 26.67 gadgets with a profit of 2800, but we cannot sell two-thirds of a gadget, and simply rounding down loses money. Finding the best whole-number plan is exactly what makes integer programming powerful - and hard!

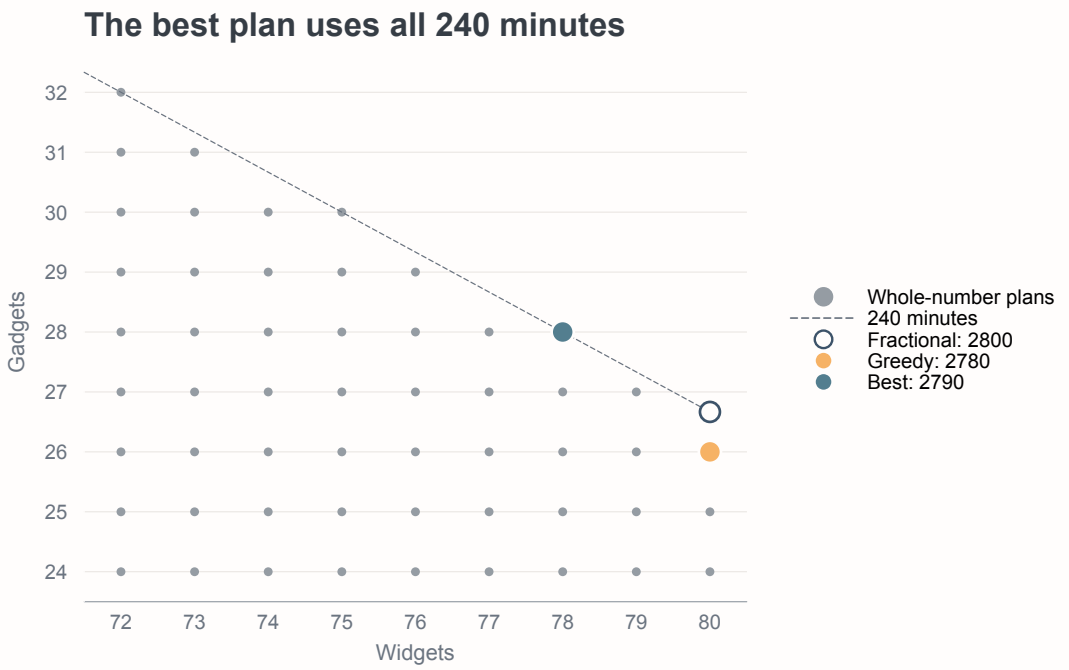
A plot makes this easier to see. It shows the whole-number plans near the best one and the line where all 240 minutes are used.

The next cell sets the look of the plots below: the course colors, quiet grey labels and no frame. Run it once. A plot picks it up when you add `course_style...` to its keywords, after a semicolon.

```
using Plots
# The course look for plots, used like this: bar(x, y; course_style...)
course_style = (
    palette=["#537E8F", "#F6B265", "#3D5369", "#A7C7C6", "#DB6B6B", "#FCCF9C"],
    background_color="#FFFDFC",
    fontfamily="Helvetica",
    titlefontfamily="Helvetica Bold",
    titlefontsize=18,
    titlefontcolor="#363D45",
    titlelocation=:left,
    foreground_color_text="#6B7480",
    foreground_color_guide="#6B7480",
```

```
    foreground_color_axis="#9AA1A8",
    foreground_color_border="#9AA1A8",
    yforeground_color_border="#FFFDFC", # no line on the left, the grid is
enough
    guidefontsize=12,
    tickfontsize=11,
    legendfontsize=11,
    foreground_color_legend=nothing,
    background_color_legend=nothing,
    grid=:y,
    gridcolor="#E8E4E0",
    gridalpha=1,
    framestyle=:axes,
    tick_direction=:none,
    yformatter=:plain,
    dpi=300,
    top_margin=4Plots.mm,
    left_margin=8Plots.mm,
    bottom_margin=6Plots.mm
)


# All whole-number plans near the best ones that fit into the 240 minutes
plans = [(w, g) for w in 72:80 for g in 24:32 if 2w + 3g <= 240]
p = scatter(first.(plans), last.(plans); course_style...,
    color="#959CA3", markersize=4, markerstrokewidth=0, label="Whole-number
plans",
    title="The best plan uses all 240 minutes", xlabel="Widgets",
ylabel="Gadgets",
    xticks=72:80, yticks=24:32, xlims=(71.5, 80.5), ylims=(23.5, 32.5),
    legend=:outerright, size=(900, 550))
plot!(p, w -> (240 - 2w) / 3, 71.5, 80; color="#6B7480", linestyle=:dash,
label="240 minutes")
scatter!(p, [80], [80 / 3]; color=course_style.background_color,
markerstrokecolor="#3D5369",
    markerstrokewidth=2, markersize=9, label="Fractional: 2800")
scatter!(p, [80], [26]; color="#F6B265",
markerstrokecolor=course_style.background_color,
    markerstrokewidth=2, markersize=10, label="Greedy: 2780")
scatter!(p, [value(widgets)], [value(gadgets)]; color="#537E8F",
markerstrokecolor=course_style.background_color,
    markerstrokewidth=2, markersize=10, label="Best: $(round(Int,
objective_value(model)))")
```



> **Note**
> Why did we tighten the MIP gap to 0 before solving? With the 1% gap from Section 1 still active, the solver is allowed to stop at any plan it can prove to be within 1% of the best. When we tried it, HiGHS stopped at 79 widgets and 27 gadgets with a profit of 2785 - "good enough" by the rule we gave it, but not the true optimum. Keep this in mind whenever you combine gap tolerances with tests that expect exact values.

Exercise 2.1 - Modify and Solve the Problem

Now it's your turn! Modify the problem above by:

1. Changing the production time constraint to 300 minutes
2. Increasing the profit for widgets to 30
3. Solving the modified problem and printing the results

> **Tip**
> Re-initializing with `Model(HiGHS.Optimizer)` gives you a completely fresh model. Remember from Section 1 that solver options belong to the model - the time limit, MIP gap, and presolve settings you set earlier are gone, and you would have to set them again if you need them.

```
# YOUR CODE BELOW
# Hint: Copy the code above and make the necessary changes
model = Model(HiGHS.Optimizer) # Don't forget to re-initialize the model


# Test your answer
@assert termination_status(model) == MOI.OPTIMAL "The termination status should
```

```
be OPTIMAL but is $(termination_status(model))"
@assert isapprox(objective_value(model), 3780, atol=1e-6) "The objective value
should be 3780 but is $(objective_value(model))"
println("Excellent work! You've successfully modified and solved the
optimization problem.")
```

> **Note**
> Why does the test only check the profit and not the number of widgets and gadgets? Because the modified problem has several optimal plans: 80 widgets and 46 gadgets, 79 and 47, and 78 and 48 all respect the time limit and earn exactly 3780. Both products now bring in the same profit per unit, so the solver is free to return any of these plans - which one you get depends on the solver's internal path. Whenever a problem can have several optimal solutions, test the objective value, not the variable values.

Section 3 - Interpreting Solver Output

When we solve an optimization problem, the solver gives us information about how it went. Let's look at some key pieces of information for the problem we solved in Section 2:

```
println("Termination status: ", termination_status(model))
println("Primal status: ", primal_status(model))
println("Dual status: ", dual_status(model))
println("Objective value: ", objective_value(model))
println("Solve time: ", solve_time(model))
```

```
Termination status: OPTIMAL
Primal status: FEASIBLE_POINT
Dual status: NO_SOLUTION
Objective value: 2790.0
Solve time: 0.0048976250109262764
```

Let's break this down:

- **Termination status**: Tells if the solver found an optimal solution, ran out of time, etc.
- **Primal status**: Indicates if we have a valid solution for our original problem
- **Dual status**: Relates to the mathematical properties of the solution; it becomes important for sensitivity analysis later in the course. For integer problems like ours there is no dual solution, which is why it shows `NO_SOLUTION` here.
- **Objective value**: The value of our objective function (in this case, our profit)
- **Solve time**: How long it took to solve the problem

Conclusion

Well done! You've completed the tutorial on advanced solver options with HiGHS in JuMP. You've learned how to set advanced solver options. Continue to the next file to learn more.

Solutions

You will likely find solutions to most exercises online. However, I strongly encourage you to work on these exercises independently without searching explicitly for the exact answers to the exercises. Understanding someone else's solution is very different from developing your own. Use the lecture notes and try to solve the exercises on your own. This approach will significantly enhance your learning and problem-solving skills.

Remember, the goal is not just to complete the exercises, but to understand the concepts and improve your programming abilities. If you encounter difficulties, review the lecture materials, experiment with different approaches, and don't hesitate to ask for clarification during class discussions.

Bibliography